

4.1.12 EEI4360 Introduction to Artificial Intelligence

Level (semester)	4 (3)					
Course Code	EEI4360					
Course Title	Introduction to Artificial Intelligence					
Credit value	3					
Core/Optional	Compulsory					
Prerequisites	(MHZ3356(CA) or MHZ4359(CA)), EEI3372(CA)					
Hourly Breakdown	Theory		Activity* hours	Independent Learning	Assessments	Total hours
	21 SSS*2h =42h	6 ILS*2h =12h	3 LAB * 3h = 9 h 3 LAB-PS * 3h = 9 h 2 HA * 4h =8h	21 SSS * 3 h =66 h 3 LAB * 1.5h = 4.5 h	1 CAT * 1.25 h =1.25 h 1 OQ*1h = 1h 1 Lab-EV *0.5 h = 0.5 h	153.2 5h
	*Practical/Design/Field work/case study					
Course Aim/s	To provide a foundation for understanding intelligent systems and algorithms, to explore basic AI techniques and tools for problem-solving and encourage ethical considerations in AI development.					
Program Learning Outcomes (PLO) addressed by course	PLO1: Apply knowledge of software engineering principles, fundamentals of computer science and mathematics, to the abstraction and conceptualization of computing models from defined problems and requirements. PLO3: Design and evaluate solutions for complex computing problems, and design and evaluate systems, components or processes that meet specified needs. PLO4: Create, select and apply appropriate techniques, resources and tools to software engineering activities. PLO8: Demonstrate knowledge and understanding of societal, health, safety, legal, and cultural issues within local and global contexts in formulating solutions to problems. PLO9: Practice software engineering profession adhering to ethical responsibilities and professional standards. PLO10: Engage in independent and lifelong learning in the broad context of technological change.					

Course Learning Outcomes (CLO):	At the completion of this course students will be able to, CLO1: Explain the principles and goals of AI. [PLO1] CLO2: Select appropriate searching algorithms to solve a problem. [PLO1] [PLO3] [PLO4] CLO3: Implement simple AI models in a programming language. [PLO3] [PLO4] CLO4: Analyze the ethical and societal implications of AI systems [PLO8] [PLO9] [PLO10]
Content (Main topics, subtopics)	Outline Syllabus: UNIT 01: Introduction to AI Session 01: History and evolution of AI, AI schools of thought Branches of AI (Machine Learning, NLP, Robotics, etc.) and applications in real-world domains. UNIT 2: Problem Solving and Search Techniques Session 01: State-space representation and game playing Session 02: Uninformed search algorithms: BFS, DFS Session 03: Informed search algorithms: A*, Greedy, Best-First Search Session 04: Constraint satisfaction problems UNIT 3: Knowledge Representation and Reasoning Session 01: Intro to Knowledge Representation and Reasoning Session 02: Propositional logic Session 03: Predicate Logic -I Session 04: Predicate Logic -II Session 05: Reasoning mechanisms Session 06: Ontologies Session 07: Rule-based systems and Inference mechanisms UNIT 04: Machine Learning Fundamentals Session 01: Overview of supervised, unsupervised, and reinforcement learning Session 02: Regression (Predictions) Session 03: Classifications (Decision trees) Session 04: Basics of neural networks Session 05: Agent Technology UNIT 05: Ethics and Future of AI Session 01: Bias, fairness, and accountability. Session 02: Societal impacts of AI, AI trends and career opportunities. Laboratory work:

	1) Overview of AI programming tools. (e.g., Prolog, Python, NumPy, pandas, scikit-learn) 2) Solve a given problem using a search algorithm or a classification technique. 3) Use a classification or a clustering technique to solve a problem. Recommended Delivery 1 LAB will be evaluated	
Teaching-Learning methods	a) <i>Course material in print - 21 sessions.</i> b) <i>Compulsory contact sessions</i> ▪ <i>TMA evaluation</i> ▪ <i>Mini Project with evaluation</i> c) <i>Self-learning/independent learning</i> ▪ <i>Learning the course material (print, online)</i> ▪ <i>recommended reading material</i> ▪ <i>Preparations for lab work/Lab practice sessions</i> ▪ <i>Presentation for Presentation/Viva</i> d) <i>Non-compulsory contact sessions</i> ▪ <i>Interactive Learning Sessions (discussion classes)</i> ▪ <i>Laboratory classes</i>	
Assessment Strategy	Overall Continuous Assessment Mark, X 100%	Final Examination Mark, Y 100%
	Details: Tutor Marked Assignments (TMA) Continuous Assessment Test (CAT) Mini Project (MP) $X = CAT * 0.4 + OQ * 0.2 + Lab * 0.4$ If $Lab \geq 40$ $X = 0$; If $Lab < 40$	Details: Final written examination, 3h (FWE) $Y = FWE$, if $X > 40$
	Overall Assessment Mark = Z $Z = 0.5X + 0.5Y$, if $Y \geq 40$ $Z = Y$, if $Y < 40$ Condition for Pass: $Z \geq 40$	
Recommended Reading	1. Rich, E., Knight, K., Artificial Intelligence, Tata-McGraw Hill, 1991 2. Russel, S., Norvig, P., Artificial Intelligence: A Modern Approach, Pearson Education, 4th Ed, 2020 3. Sebastian R. and Mirjalili V., Python Machine Learning, 2019	